Possible restrictions on broadband total absorption in the 2d limit through sum rules

The focus of these notes is to determine if sum rules put a restriction on the bandwidth of perfect CPA for a monolayer system. We know that

$$\varepsilon(\omega)=1+\chi(\omega)=1+\frac{i\sigma(\omega)}{\omega\varepsilon_0}\to\sigma(\omega)=-i\omega\varepsilon_0\chi(\omega)$$

From this, we obtain: $\Re\sigma(\omega)=\omega\varepsilon_0\Im\chi(\omega),\Im\sigma(\omega)=-\omega\varepsilon_0\Re\chi(\omega)$

Using the sum rule, https://hackmd.io/@aligho/Bkk_HLUwlg (https://hackmd.io/@aligho/Bkk_HLUwlg),

$$\Re\chi(\omega)=\frac{2}{\pi}\int_0^\infty\frac{\omega'\Im\chi(\omega')}{\omega'^2-\omega^2}\mathrm{d}\omega'$$

We obtain (setting $\sigma(\omega)=\sigma'(\omega)+i\sigma''(\omega)$):

$$\lim_{\omega\to\infty}\omega\sigma''(\omega)=\frac{2}{\pi}\int_0^\infty\sigma'(\omega')\mathrm{d}\omega'$$

If we assume that $\sigma''(\omega)\approx ne^2/m\omega$ at high frequencies (and use the CPA condition for double-sided illumination of a monolayer, $\sigma(\omega)=2c\varepsilon_0$), we have that:

$$\frac{\pi ne^2}{2m}>2c\varepsilon_0(\Delta_\omega)\to\frac{\pi ne^2}{4mc\varepsilon_0}=\left(\frac{e^2}{4\pi\varepsilon_0\hbar c}\right)\frac{\pi^2 n\hbar}{m}=\alpha\pi^2 n\hbar/m>\Delta_\omega$$

Therefore, the bound is linear in the electronic density. We estimate this:

$$\begin{aligned}\frac{\pi^2}{137}\frac{n}{0.5\times10^6\mathrm{eV}}(9\times10^{20}\mathrm{cm}^2)(6.67\times10^{-16}\mathrm{eV})\times10^{-12}\mathrm{THz}\approx\\(0.072)(18n)\times10^{-14}\mathrm{cm}^2\mathrm{THz}=1.3n\times10^{-14}\mathrm{cm}^2\mathrm{THz}\end{aligned}$$

Therefore, for a reasonable electron density, we can have broadband CPA over a band of at most a few terahertz.